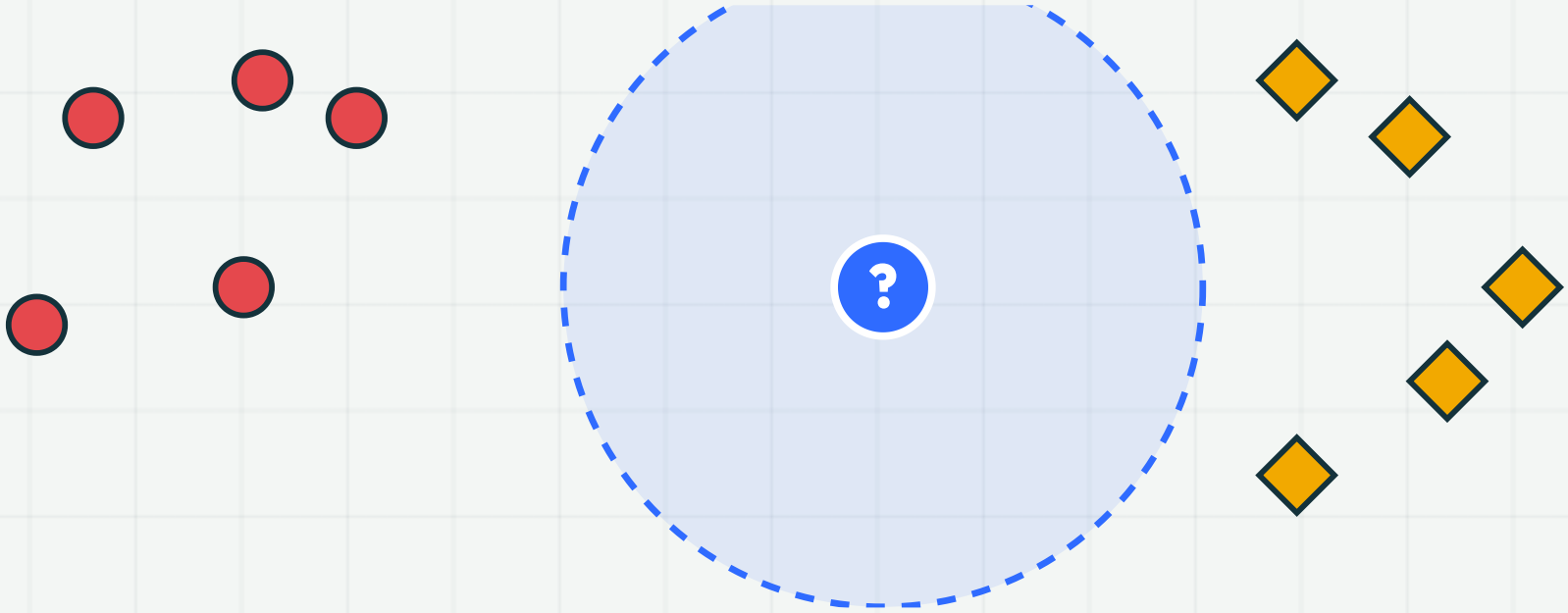
DAY 181/365 · KNN · PART 1 OF 3

Tell me who your neighbours are

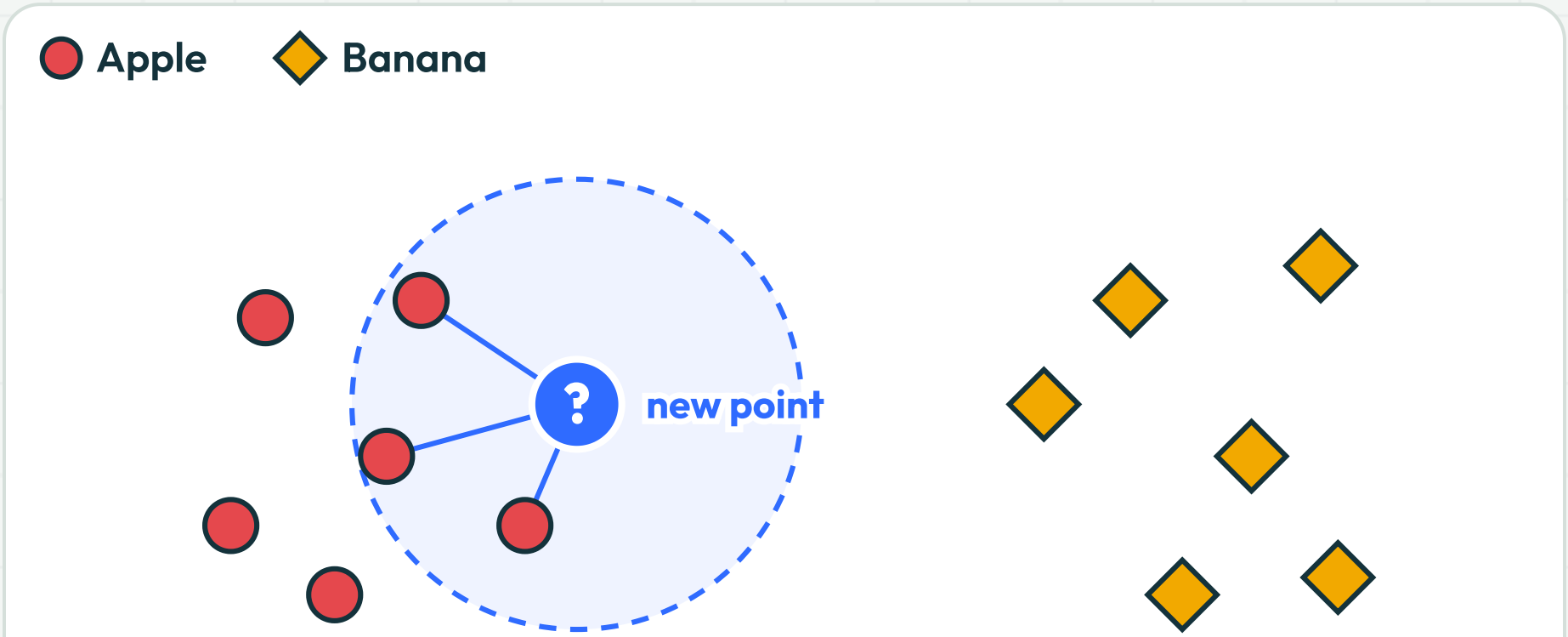
K-Nearest Neighbours, explained with pictures.



Suyog Yadav

Swipe →

The whole idea in one picture

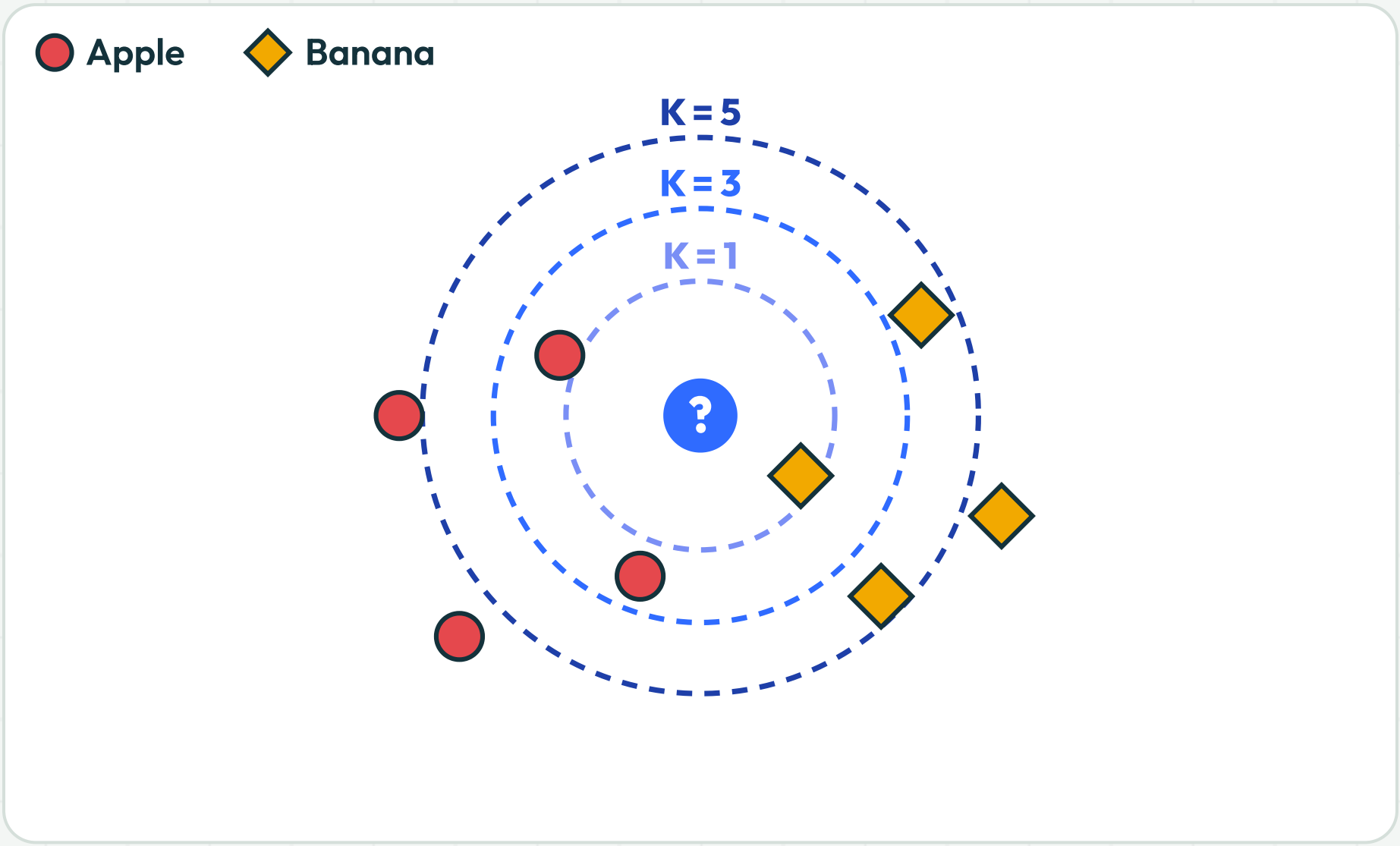


A new point arrives and we don't know what it is. KNN looks at the points **closest to it** and copies what most of them are.

Here: the 3 nearest points are all apples, so the new point is an **apple**.

What is the "K"?

K is how many neighbours we ask. The same new point, three different K values:



K = 1	K = 3	K = 5
Banana	Apple	Banana

The answer changes with K, so picking K matters. We'll tackle that in Part 2.

How KNN works: 4 steps

1 **Pick K**
For example, 3 neighbours.

↓

2 **Measure the distance**
From the new point to every point we've stored.

↓

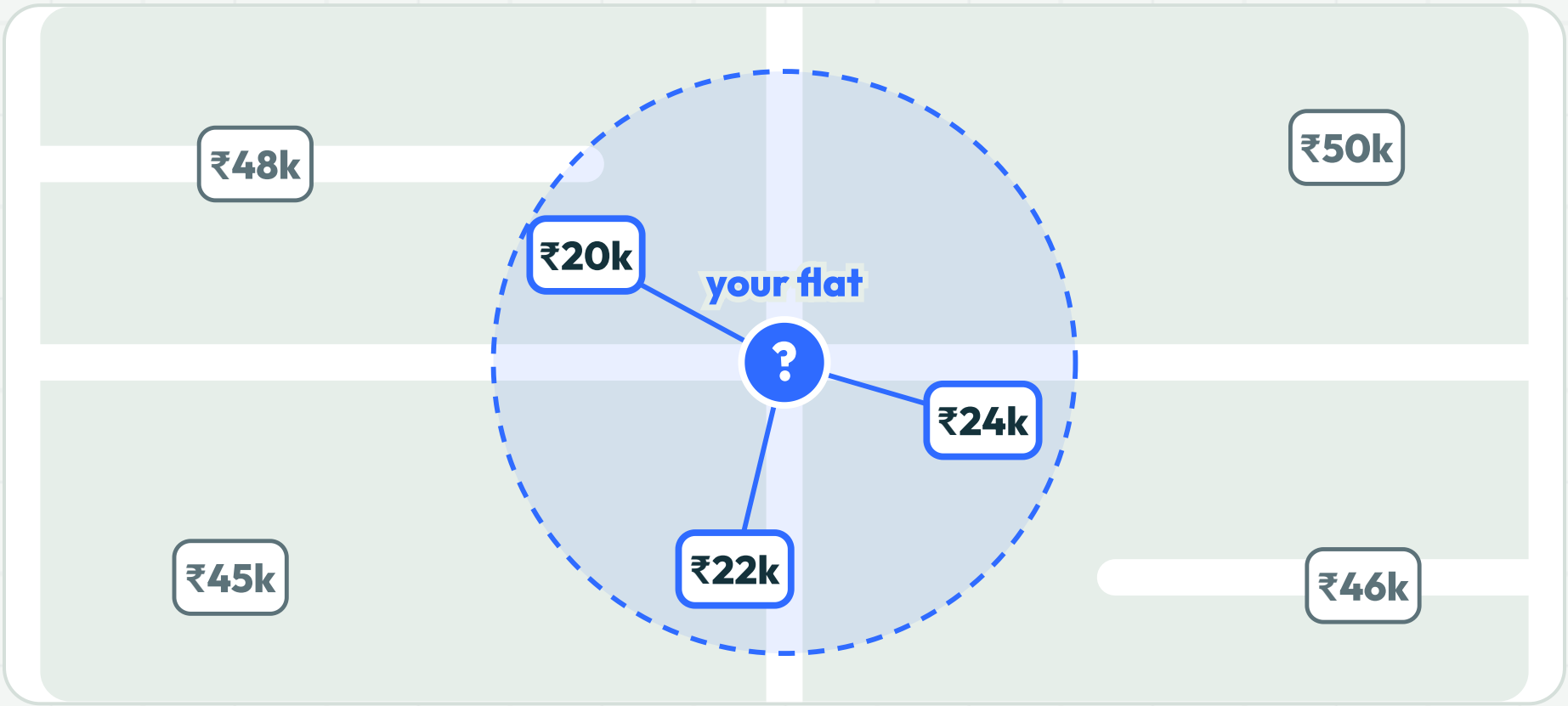
3 **Take the K closest**
Sort by distance and keep the top K.

↓

4 **Decide**
Vote for a category, or average for a number.

A vote for categories, an average for numbers

Predicting a number: what rent should your flat charge? Look at the 3 closest flats.



(20 + 22 + 24) ÷ 3 = ₹22k

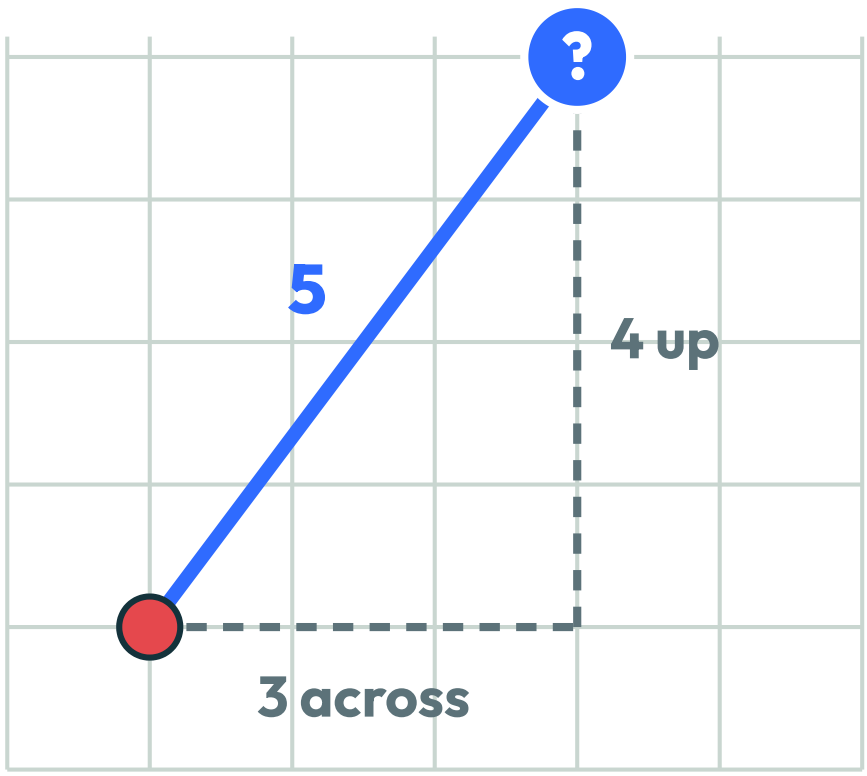
The far-away flats cost more, but they're not neighbours, so they're ignored.

Illustrative rents.

Category → **vote** (apple or banana). **Number** → **average** (rent).

"Closest" means the shortest distance

To find neighbours, KNN needs a way to measure how far apart two points are. The simplest is the **straight line**.



From (1, 1) to (4, 5): go **3 across** and **4 up**. The straight-line distance is **5**, since $3^2 + 4^2 = 25$ and $\sqrt{25} = 5$.

There are other ways to measure distance. That's Part 2.

KNN never studies. It just remembers.

Most models

Study the data first (training) | Answer | fast

KNN

Just store it | Answer: search every stored point

Sketch of where the effort goes, not a measurement.

A "lazy learner"

There's no training step. KNN stores all the examples and only does the work **when you ask a question**.

Like an open-book exam

It doesn't learn a rule. It looks things up each time.

TAKEAWAYS

To label a new point, ask its K nearest neighbours.

Category: take a vote. **Number:** take an average.

K is how many neighbours you ask, and it can change the answer.

No training: KNN stores the data and searches it when asked.

Part 2: how to measure distance (Euclidean, Manhattan) and how to choose K.